



Can a quadruped robot climb ladders?

A team of robotics engineers from the Robotics Systems Lab at ETH Zurich have been able to modify a quadruped robot, ANYmal, to climb standard ladders. They found that typical robotic ‘hands’ or ‘paws’ were not suitable for



Composite image of a quadrupedal robot equipped with hooked end-effectors, ascending a ladder in 4 seconds with a reinforcement learning-based control policy. Ladder shown has parameters 90° incline, 1.8m length, 0.3m inter-rung spacing, 2.5cm rung radius, and 1m width (Credit: arXiv, 2024)

climbing ladders, so they developed a robotic paw that forms a hook-like grip, which can tighten around rungs. They used reinforcement learning, facilitated by a simulation with a teacher-student method, where the ‘teacher’ robot accessed videos of various ladder-climbing scenarios.

This method improved the robot’s climbing capabilities, allowing ‘student’ robots to learn by imitating the teacher. After training, the modified robot achieved a success rate of about 90% in real-world ladder climbing tests, outperforming counterparts without the specialised hooked paws.

Solar-powered desalination system

MIT researchers have developed a solar-powered desalination system that adjusts to sunlight intensity without the need for batteries, reducing costs and complexity. The system speeds up or slows down desalination based on available solar energy, maximising efficiency. Tested in New

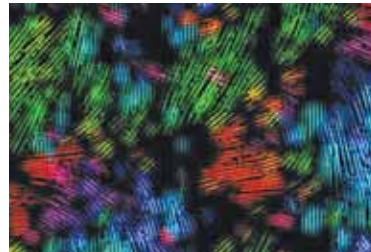


MIT engineers tested a community-scale prototype on groundwater wells in New Mexico in the US over six months (Photo: Shane Pratt)

Mexico, a prototype processed up to 5000 litres of water daily, using over 94% of the solar energy generated, even with varying sunlight. The system targets brackish groundwater, abundant in areas with scarce freshwater. Designed for off-grid communities, it offers a sustainable solution for clean water, with plans for scaling up and launching a company based on this innovation.

How do new soft semiconductors work?

Scientists at Stanford have applied a specialised electron microscopy technique for soft, ‘beam-sensitive’ materials to explore the structural dynamics of organic mixed ionic-electronic conductors (OMIECs), revealing how their favourable electro-



Representative map visualises the direction of polymer chains in edge-on crystallites (Credit: Salleo Research Group)

chemical properties function. OMIECs incorporate a liquid electrolyte between polymer layers, like water in a car battery, facilitating ion movement and electrical current generation. This technique, which has traditionally struggled to visualise

such materials without damage, showed how the polymer maintains integrity while expanding and how it stretches around electrolyte bubbles to form a continuous electronic path. The study used Cryo 4D-STEM to freeze and observe the material in a vitrified state, noting minimal structural changes in the polymer chains during charging and discharging, leading to efficient ion exchange with minimal strain.

Design for more efficient EV batteries

Researchers at McGill University have made a significant breakthrough in all-solid-state lithium battery technology, addressing a key challenge in electric vehicle (EV) batteries. A major issue has been the high resistance at the interface between the ceramic electrolyte and electrodes, which limits energy efficiency. The team developed a porous ceramic membrane filled with a small amount of polymer, reducing this resistance and allowing lithium ions to move freely. This innovation improves battery performance, enabling safer, longer lasting, and more efficient EV batteries. It represents a major step toward replacing flammable liquid electrolytes with safer solid-state alternatives.

Rapid 3D mapping drones

Researchers from Sun Yat-Sen University and the Hong Kong University of Science and Technology have developed SOAR, a new drone system for autonomous exploration and 3D reconstruction of real-world environments. It uses a fleet of UAVs equipped with lidar and visual sensors. The system’s key innovation combines fast scene exploration with high-quality 3D mapping. One drone map is using lidar, while others capture visual data with viewpoints optimised in real-time. SOAR’s novel algorithms improve coordination and data collection, outperforming current methods in speed and accuracy.